

Aligned Dipole-Dipole Interactions

Starting with the dipole interactions

$$V_{dd}(\mathbf{R}) = \frac{1}{R^3} - 3 \frac{(\mathbf{R} \cdot \hat{\mathbf{d}})^2}{R^5},$$

with $\mathbf{R} = \mathbf{r}_i - \mathbf{r}_j$. The first derivative reads

$$\frac{\partial V_{dd}}{\partial R_\alpha} = -\frac{3R_\alpha}{R^5} + \frac{15R_\alpha(\mathbf{R} \cdot \hat{\mathbf{d}})^2}{R^7} - \frac{6(\mathbf{R} \cdot \hat{\mathbf{d}})d_\alpha}{R^5},$$

the second derivative reads consequently

$$\begin{aligned} \frac{\partial^2 V_{dd}}{\partial R_\alpha \partial R_\beta} &= -3 \frac{\delta_{\alpha\beta}}{R^5} + 15 \frac{R_\alpha R_\beta}{R^7} + 15 \frac{\delta_{\alpha\beta}(\mathbf{R} \cdot \hat{\mathbf{d}})^2}{R^7} - 15 \cdot 7 \frac{R_\alpha R_\beta (\mathbf{R} \cdot \hat{\mathbf{d}})^2}{R^9} \\ &\quad + 30 \frac{d_\alpha R_\beta + d_\beta R_\alpha}{R^7} (\mathbf{R} \cdot \hat{\mathbf{d}}) - 6 \frac{d_\alpha d_\beta}{R^5} \\ &= \delta_{\alpha\beta} \left(-3 \frac{1}{R^5} + 15 \frac{(\mathbf{R} \cdot \hat{\mathbf{d}})^2}{R^7} \right) - 6 \frac{d_\alpha d_\beta}{R^5} + 30 \frac{d_\alpha R_\beta + d_\beta R_\alpha}{R^7} (\mathbf{R} \cdot \hat{\mathbf{d}}) \\ &\quad + 15 R_\alpha R_\beta \left(\frac{1}{R^7} - 7 \frac{(\mathbf{R} \cdot \hat{\mathbf{d}})^2}{R^9} \right). \end{aligned}$$

General Dipole-Dipole Interactions

In the case where not all dipole moments are aligned, the properties of the interaction get extended to

$$V_{dd}(\mathbf{R}) = \frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{R^3} - 3 \frac{(\mathbf{R} \cdot \hat{\mathbf{d}}_1)(\mathbf{R} \cdot \hat{\mathbf{d}}_2)}{R^5}.$$

The first derivative reads

$$\frac{\partial V_{dd}}{\partial R_\alpha} = -\frac{3(\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2) R_\alpha}{R^5} + \frac{15R_\alpha(\mathbf{R} \cdot \hat{\mathbf{d}}_1)(\mathbf{R} \cdot \hat{\mathbf{d}}_2)}{R^7} - \frac{3 \left((\mathbf{R} \cdot \hat{\mathbf{d}}_1) d_{2,\alpha} + (\mathbf{R} \cdot \hat{\mathbf{d}}_2) d_{1,\alpha} \right)}{R^5},$$

the second derivative reads consequently

$$\begin{aligned} \frac{\partial^2 V_{dd}}{\partial R_\alpha \partial R_\beta} &= -3 \frac{\delta_{\alpha\beta}(\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2)}{R^5} + 15 \frac{R_\alpha R_\beta (\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2)}{R^7} + 15 \frac{\delta_{\alpha\beta}(\mathbf{R} \cdot \hat{\mathbf{d}}_1)(\mathbf{R} \cdot \hat{\mathbf{d}}_2)}{R^7} - 15 \cdot 7 \frac{R_\alpha R_\beta (\mathbf{R} \cdot \hat{\mathbf{d}}_1)(\mathbf{R} \cdot \hat{\mathbf{d}}_2)}{R^9} \\ &\quad + 15 \left[\frac{d_{1,\alpha} R_\beta + d_{1,\beta} R_\alpha}{R^7} (\mathbf{R} \cdot \hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} R_\beta + d_{2,\beta} R_\alpha}{R^7} (\mathbf{R} \cdot \hat{\mathbf{d}}_1) \right] - 3 \frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{R^5} \\ &= \delta_{\alpha\beta} \left(-3 \frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{R^5} + 15 \frac{(\mathbf{R} \cdot \hat{\mathbf{d}}_1)(\mathbf{R} \cdot \hat{\mathbf{d}}_2)}{R^7} \right) - 3 \frac{d_{1,\alpha} d_{2,\beta} + d_{2,\alpha} d_{1,\beta}}{R^5} \\ &\quad + 15 \left[\frac{d_{1,\alpha} R_\beta + d_{1,\beta} R_\alpha}{R^7} (\mathbf{R} \cdot \hat{\mathbf{d}}_2) + \frac{d_{2,\alpha} R_\beta + d_{2,\beta} R_\alpha}{R^7} (\mathbf{R} \cdot \hat{\mathbf{d}}_1) \right] \\ &\quad + 15 R_\alpha R_\beta \left(\frac{\hat{\mathbf{d}}_1 \cdot \hat{\mathbf{d}}_2}{R^7} - 7 \frac{(\mathbf{R} \cdot \hat{\mathbf{d}}_1)(\mathbf{R} \cdot \hat{\mathbf{d}}_2)}{R^9} \right). \end{aligned}$$

For the the Van der Waals interaction the potential has to be squared. Therefore the second derivative reads

$$\begin{aligned} \frac{\partial^2 V_{dd}^2}{\partial R_\alpha^2} &= 2 V_{dd} \frac{\partial^2 V_{dd}}{\partial R_\alpha^2} \\ \frac{\partial^2 V_{dd}^2}{\partial R_\alpha \partial R_\beta} &= 2 \frac{\partial V_{dd}}{\partial R_\alpha} \frac{\partial V_{dd}}{\partial R_\beta} + 2 V_{dd} \frac{\partial^2 V_{dd}}{\partial R_\alpha \partial R_\beta} \end{aligned}$$

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Winding Number

A chiral Hamiltonian is defined through the property, that it anticommutes with a unitary hermitian operator Γ

$$\Gamma H \Gamma = -H \quad \text{with} \quad \Gamma^2 = 1. \quad (0.1)$$

From this property follows that the Hamiltonian can be written in block off-diagonal form

$$H = \begin{pmatrix} 0 & h \\ h^\dagger & 0 \end{pmatrix}, \quad (0.2)$$

where h is, in general, a non-Hermitian matrix. This structure also applies for the matrix in momentum space and allows the definition of a winding number, which characterizes the topological properties of the system. If the Hamiltonian is 2×2 dimensional, the winding number can be defined through the decomposition of H in terms of the Pauli matrices

$$H = \mathbf{n} \cdot \boldsymbol{\sigma}, \quad (0.3)$$

where $\mathbf{n} = (n_x, n_y, 0)$ is a vector in the xy -plane. The winding number is then given by the winding of the vector $\mathbf{n}$ around the origin when traversing the Brillouin zone. Identifying the xy -plane of $\mathbf{n}$ with the complex plane, which defines a path $\mathbf{n}(k) \rightarrow \gamma(k)$, the winding number can be calculated through

$$w = \frac{1}{2\pi i} \oint_{\gamma_k} \frac{1}{z} dz = \frac{1}{2\pi i} \int_{\text{BZ}} dk \frac{1}{\gamma(k)} \frac{d\gamma(k)}{dk}.$$

In terms of the Hamiltonian above this reads

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \frac{h'(k)}{h(k)}.$$

Generalizing the winding number to higher dimensional Hamiltonians is possible through the use of the block off-diagonal form. The winding number then reads

$$w = \frac{1}{2\pi i} \int_{\text{BZ}} dk \text{Tr} [h^{-1}(k) \partial_k h(k)].$$

Diagonalizing h as $h = S \Lambda S^{-1}$ with the diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots)$ and the invertible matrix S , the trace can be rewritten as

$$\begin{aligned}\text{Tr} [h^{-1} \partial_k h] &= \text{Tr} [S \Lambda S^{-1} \partial_k (S \Lambda S^{-1})] \\ &= \text{Tr} [\Lambda^{-1} \partial_k \Lambda + S^{-1} \partial_k S + (\partial_k S^{-1}) S] \\ \text{Tr} [S^{-1} \partial_k S + S \partial_k S^{-1}] &= \text{Tr} [\partial_k (S S^{-1})] = 0, \\ \Rightarrow \text{Tr} [h^{-1} \partial_k h] &= \text{Tr} [\Lambda^{-1} \partial_k \Lambda] = \sum_i \frac{\partial_k \lambda_i}{\lambda_i} = \partial_k \left(\sum_i \ln(\lambda_i) \right) = \partial_k \ln(\det[h]).\end{aligned}$$

Another useful definition of the winding number is through the so-called Q -matrix, which is defined through the projectors onto the eigenstates of the Hamiltonian. As the Hamiltonian is chiral, the eigenvalues come in pairs $\pm E$. We introduce a labeling of the eigenvalues, such that $E_j = -E_{-j}$ with

$j = 1, 2, \dots, N$ and N being half the dimension of the Hamiltonian. The corresponding eigenstates are labeled accordingly ($|\psi_j\rangle$) and posses the property $\Gamma |\psi_j\rangle = |\psi_{-j}\rangle$. The Q-matrix is defined as

$$Q = \sum_{j=1}^N (|\psi_j\rangle \langle \psi_j| - |\psi_{-j}\rangle \langle \psi_{-j}|).$$

As a consequence of the chiral symmetry, the Q-matrix anticommutes with Γ and can therefore be written in block off-diagonal form in the basis, where Γ is diagonal, which is also the case in which H is block off-diagonal.

$$\begin{aligned} \Gamma Q \Gamma &= \sum_{j=1}^N (|\psi_{-j}\rangle \langle \psi_{-j}| - |\psi_j\rangle \langle \psi_j|) = -Q \\ \Rightarrow Q &= \begin{pmatrix} 0 & q \\ q^\dagger & 0 \end{pmatrix} \end{aligned}$$

Labeling the eigenvalues and eigenstates of h as h_i and $|h_i\rangle$ respectively, a basis of H can be constructed that preserves the block off-diagonal structure. The basis is split into two sets A and B,

$$|h_i^A\rangle = \begin{pmatrix} |h_i\rangle \\ 0 \end{pmatrix}, \quad |h_i^B\rangle = \begin{pmatrix} 0 \\ |h_i\rangle \end{pmatrix}.$$

The eigenstates of H can be constructed from this as

$$\begin{aligned} |H_i^+\rangle &= \frac{1}{\sqrt{2}|h_i|} (|h_i\rangle |h_i^A\rangle + h_i^* |h_i^B\rangle) \\ |H_i^-\rangle &= \frac{1}{\sqrt{2}|h_i|} (|h_i\rangle |h_i^A\rangle - h_i^* |h_i^B\rangle) \\ \text{with } H |H_i^\pm\rangle &= \pm |h_i| |H_i^\pm\rangle. \end{aligned}$$

The Q-matrix can thus be written as

$$\begin{aligned} Q &= \sum_i (|H_i^+\rangle \langle H_i^+| - |H_i^-\rangle \langle H_i^-|) \\ &= \sum_i \frac{1}{|h_i|} (h_i |h_i^A\rangle \langle h_i^B| + h_i^* |h_i^B\rangle \langle h_i^A|) \end{aligned}$$

It now follows that the winding number of the q-matrix is the same as that of h

$$\begin{aligned} w_q &= \frac{1}{2\pi i} \sum_j \int_{\text{BZ}} dk \frac{\partial_k \hat{h}_j}{|\hat{h}_j|} \\ &= \frac{1}{2\pi i} \sum_j \int_{\text{BZ}} dk \frac{\partial_k h_j}{h_j} \\ &= w_h. \end{aligned}$$

where $\hat{h}_j = h_j/|h_j|$ is the normalized eigenvalue. Another equivalent expression for the winding number can be given through the eigenstates of the Hamiltonian

$$\begin{aligned} w &= \frac{1}{\pi i} \sum_j \int_{\text{BZ}} dk S_j(k) \\ \text{with } S_j(k) &= \langle \Gamma H_j^- | \partial_k | H_j^- \rangle \\ &= \langle H_j^+ | \partial_k | H_j^- \rangle \\ &= \frac{1}{2} (\langle h_j^A | \partial_k | h_j^A \rangle - \langle h_j^B | \partial_k | h_j^B \rangle) - \frac{1}{2} \frac{h_j \partial_k h_j^*}{|h_j|^2} \\ &= -\frac{1}{2} \frac{\partial_k h_j^*}{h_j^*}. \end{aligned}$$